

Environment Building

pymycobot is a Python package used for serial communication with myCobot. It supports Python2, Python3.5 and later versions.

Before using pymycobot, make sure to build a Python environment. Follow the steps below to install Python.

Installing Python



- **Python** Python's official downloading address: <https://www.python.org/downloads/>
- [Python downloading and Installation Tutorial](#) for reference only

1 Download and Installation of Python

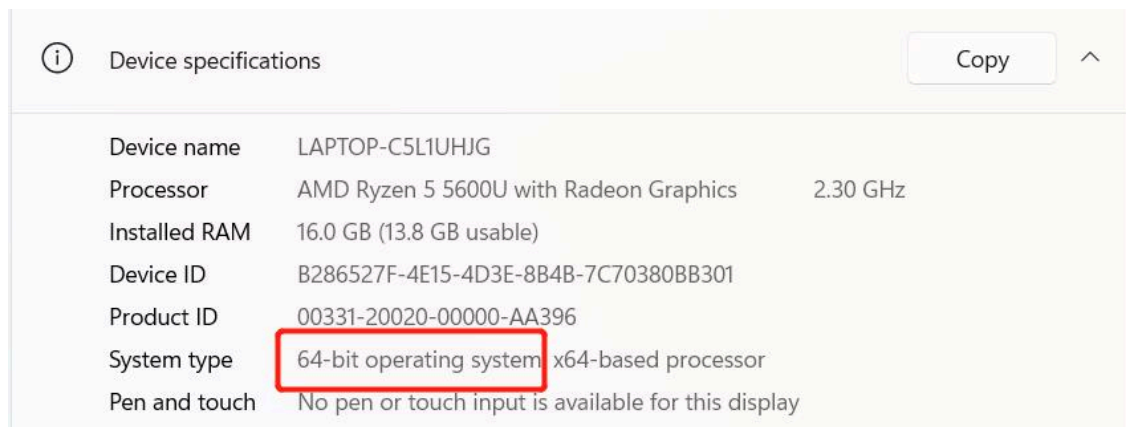
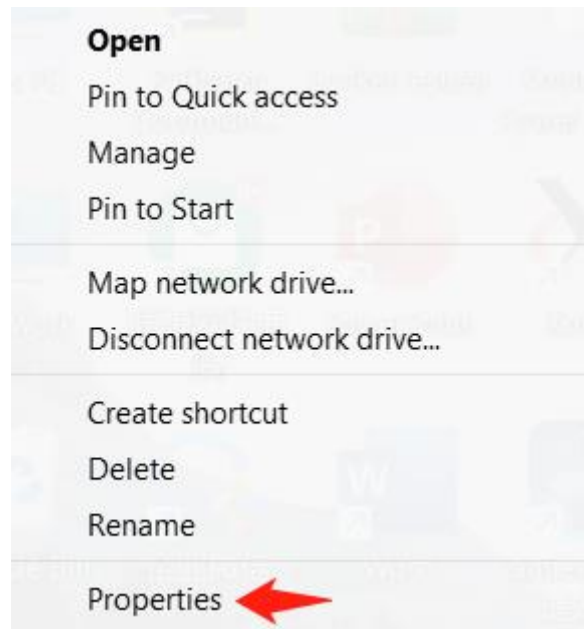
Python is applicable to:

- myCobot 280:
 - myCobot 280 M5
 - myCobot 280 PI
 - myCobot 280 Jetson Nano
 - myCobot 280 for Arduino
- myCobot 320:
 - myCobot 320 M5
 - myCobot 320 PI
- myPalletizer 260:
 - myPalletizer 260 M5
 - myPalletizer 260 PI
- mechArm-270:
 - mechArm-270 M5
 - mechArm-270 PI

At present, Python has two versions: 2.x and 3.x. These two versions are incompatible with each other. This section takes the version 3.x as an example due to its increasing popularity.

1.1 Installing Python

Notice: Before installation, check the operation system of PC. Press right button on the **My Computer** icon and then select **Properties**. Install the corresponding Python.



- Go to <http://www.python.org/download/> to download Python.
- Click on **Downloads**, and then **download begins**. Tick **Add Python 3.10 to PATH**. Click on **Install Now**, and then installation begins.

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- Download Python
- Looking for Python
- Linux/UNIX, macOS
- Want to help test d
- Docker images
- Looking for Python

All releases

Source code

Windows

macOS

Other Platforms

License

Alternative Implementations

Download for Windows

Python 3.10.7

Note that Python 3.9+ cannot be used on Windows 7 or earlier.

Not the OS you are looking for? Python can be used on many operating systems and environments.

View the full list of downloads.

Active Python Releases

For more information visit the Python Developer's Guide.

Python version	Maintenance status	First released	End of support	Release schedule
3.10	bugfix	2021-10-04	2026-10	PEP 619

<https://www.python.org/downloads/>

Python 3.10.7 (64-bit) Setup

Install Python 3.10.7 (64-bit)

Select Install Now to install Python with default settings, or choose Customize to enable or disable features.

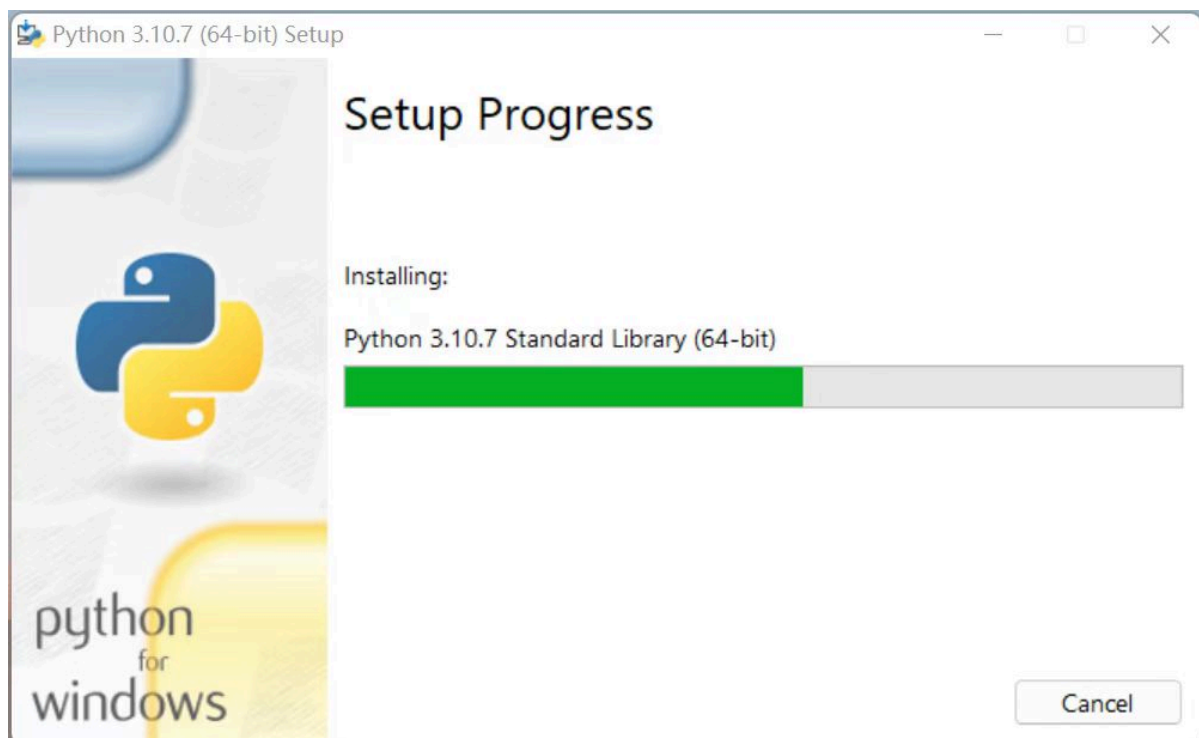
Install Now
C:\Users\Surface\AppData\Local\Programs\Python\Python310
Includes IDLE, pip and documentation
Creates shortcuts and file associations

→ Customize installation
Choose location and features

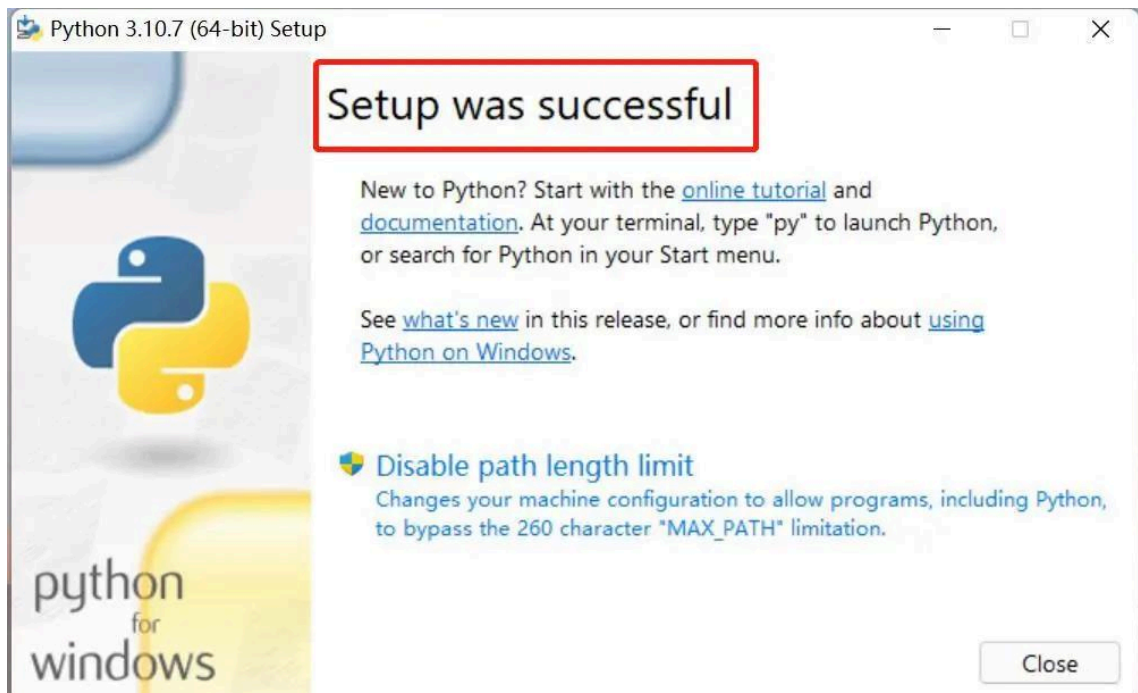
☒ Install launcher for all users (recommended)

☒ Add Python 3.10 to PATH

Cancel



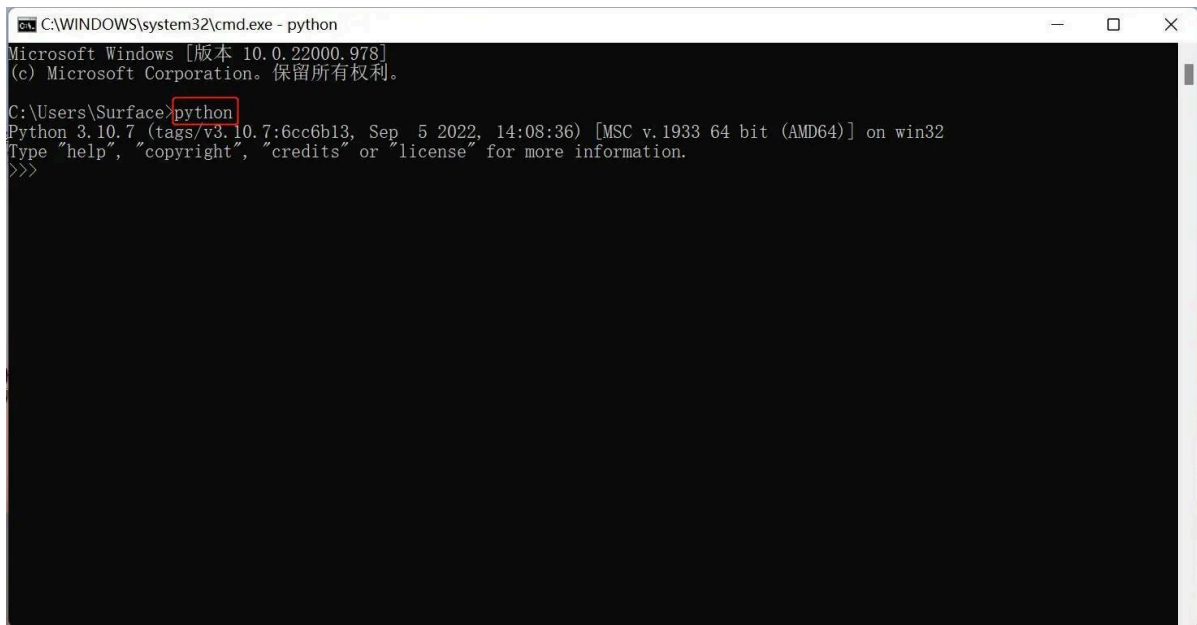
- Download and installation complete.



1.2 Running Python

Open the command prompt window (Win+R, input `cmd` and press `Enter`). Type `Python`.

Successful Installation:



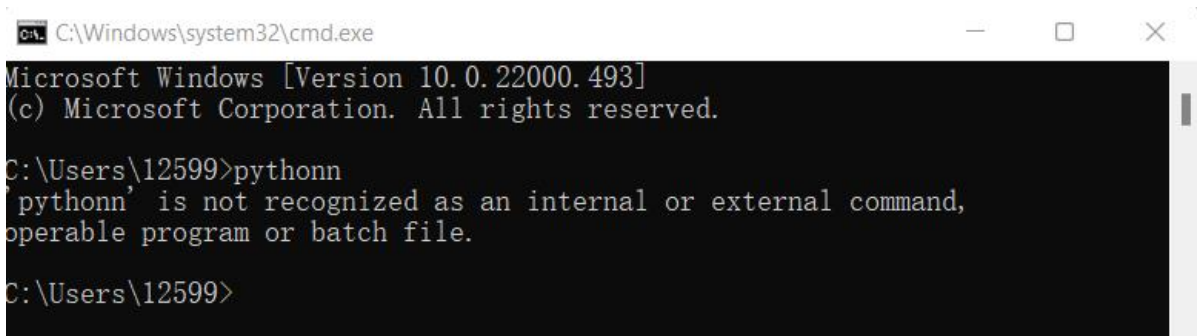
```
C:\WINDOWS\system32\cmd.exe - python
Microsoft Windows [版本 10.0.22000.978]
(c) Microsoft Corporation。保留所有权利。

C:\Users\Surface>python
Python 3.10.7 (tags/v3.10.7:6cc6b13, Sep  5 2022, 14:08:36) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

This on-screen instruction means that Python is successfully installed. The prompt `>>>` means Python interactive environment. If you input a Python code to get the execution result immediately.

Error Report:

If a wrong instruction is typed, for example "pythonn", the system may report an error.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22000.493]
(c) Microsoft Corporation. All rights reserved.

C:\Users\12599>pythonn
'pythonn' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\12599>
```

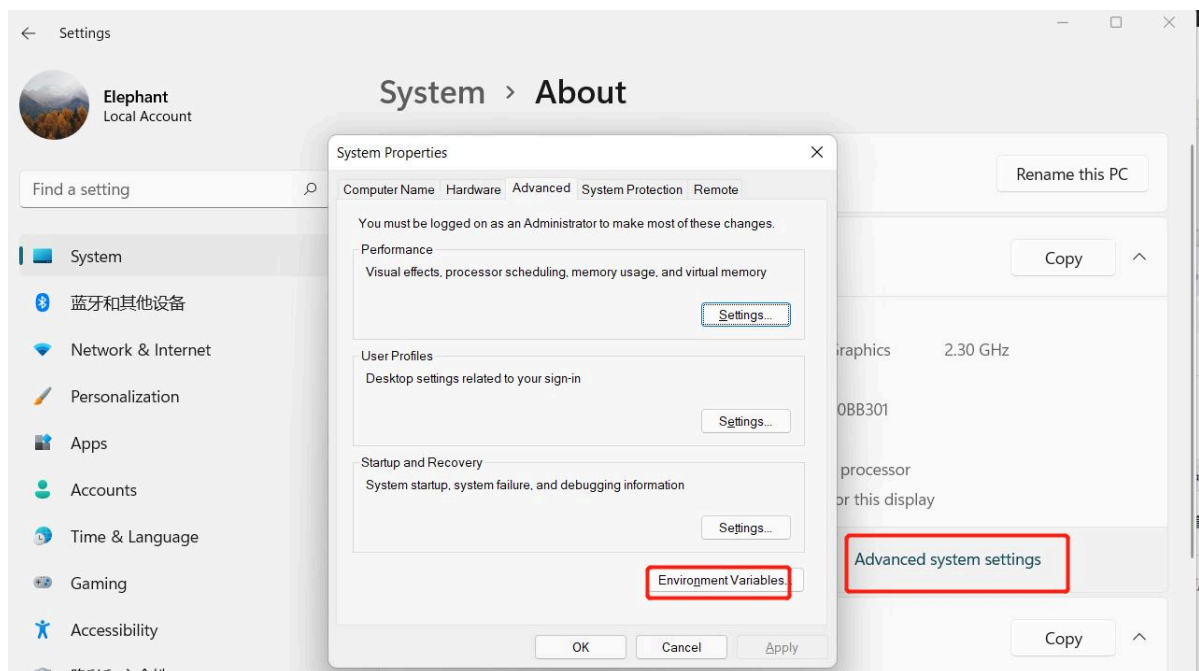
Notice: Generally, the error results from lack of environment configuration. Refer to **1.3 Environment Configuration** to solve problems.

1.3 Environment Variable Configuration

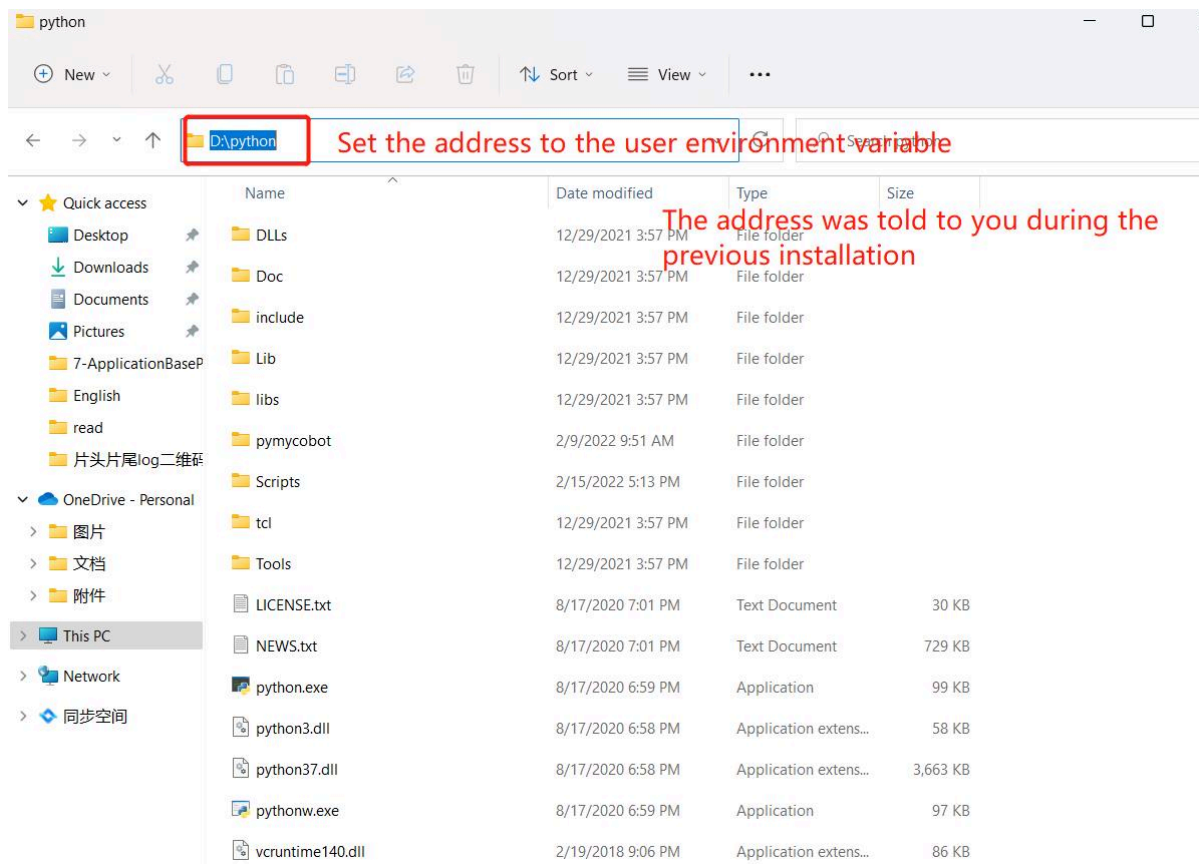
Windows follows the path set by a Path environment variable in search of **python.exe**. Otherwise, an error will be reported. If you fail to tick `Add Python 3.9 to PATH` during installation, you need to manually add the path where python.exe is located into environment variable or download python again. Remember to tick `Add Python 3.9 to PATH`.

Follow the steps below to add python into environment variable manually.

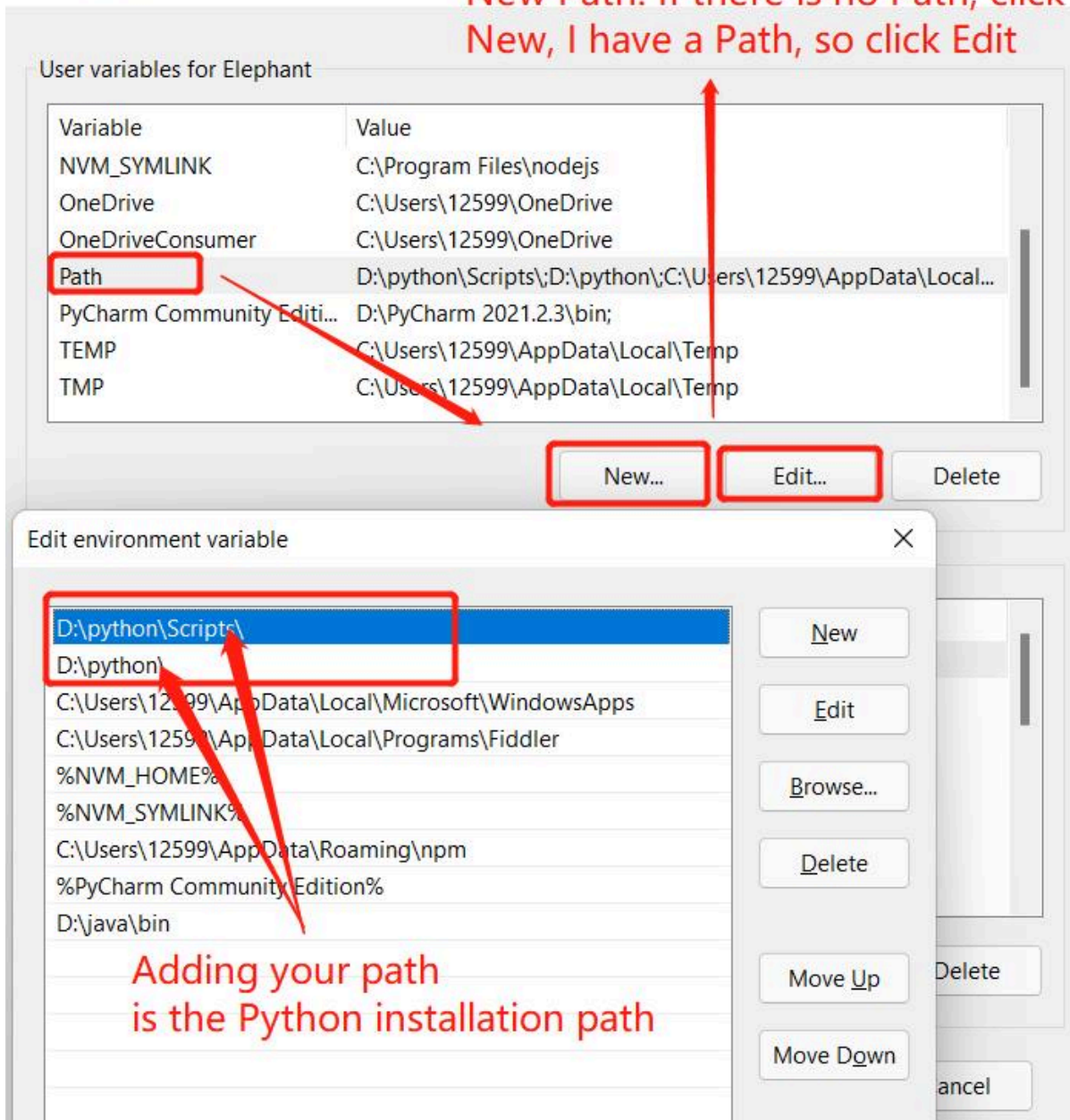
- Right click on `My Computer` icon --> Properties --> Advanced System Settings --> Environment Variables



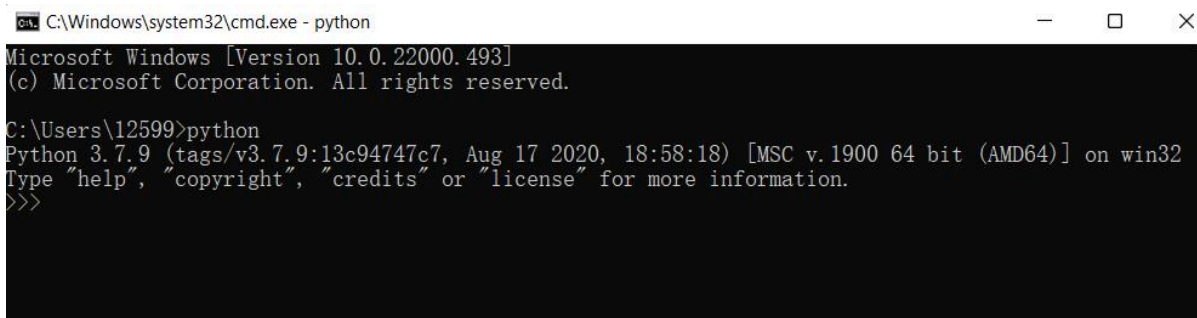
- The environment variables include user variables and system variables. For user variables, users can utilize their own downloaded programs via `cmd` command. Write the absolute path of the target program into the user variables.



New Path: If there is no Path, click New, I have a Path, so click Edit



- After the configuration, open the command prompt window (Win+R; input `cmd` and press Enter), and type `Python`.



2 Installation of PyCharm

PyCharm is a powerful python editor with the nature of cross-platform. Follow the steps below to download and install PyCharm.

Go to [PyCharm](#) to download PyCharm.

2.1 Download and Installation

Official website view:



Version: 2022.2.3
Build: 222.4345.23
11 October 2022

[System requirements](#)

[Installation instructions](#)

[Other versions](#)

[Third-party software](#)

Download PyCharm

[Windows](#)

[macOS](#)

[Linux](#)

Professional

For both Scientific and Web Python development. With HTML, JS, and SQL support.

[Download](#)

Free 30-day trial available

Community

For pure Python development

[Download](#)

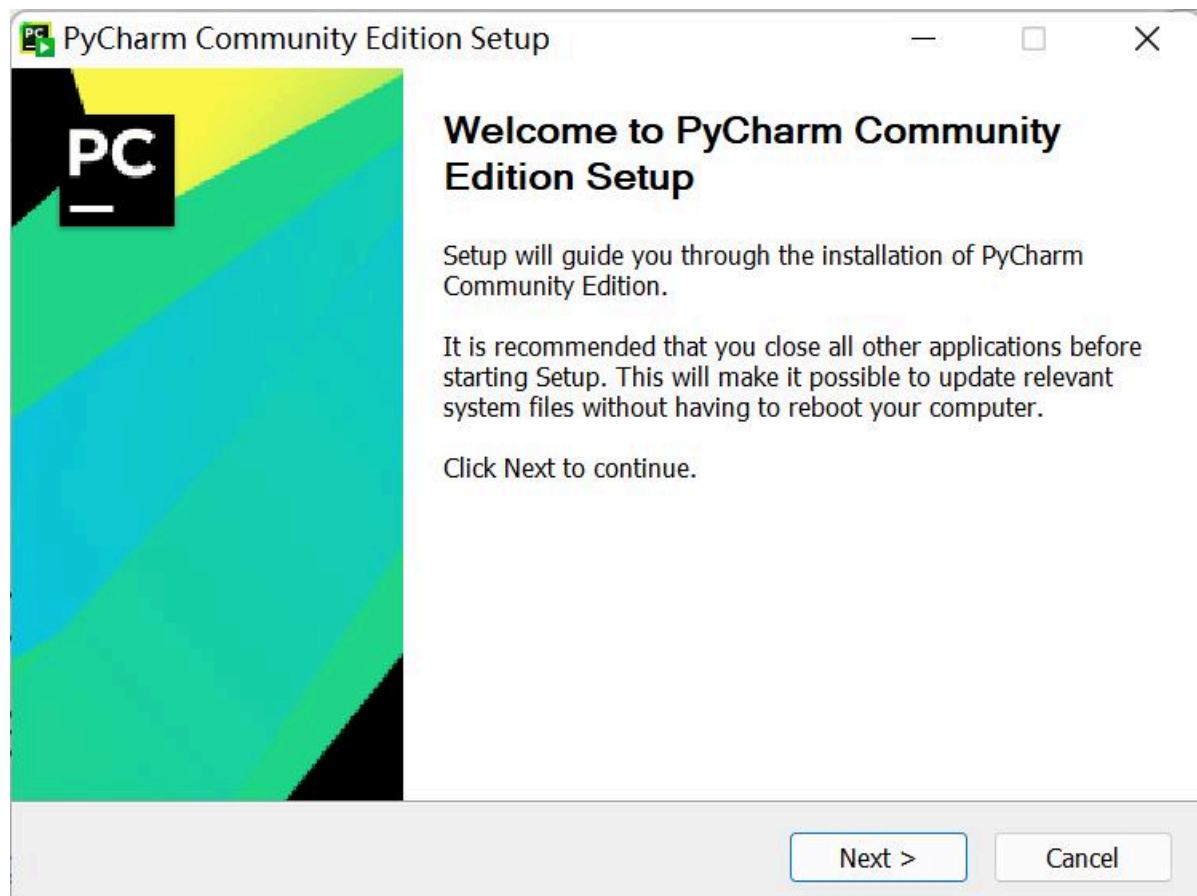
Free, built on open-source



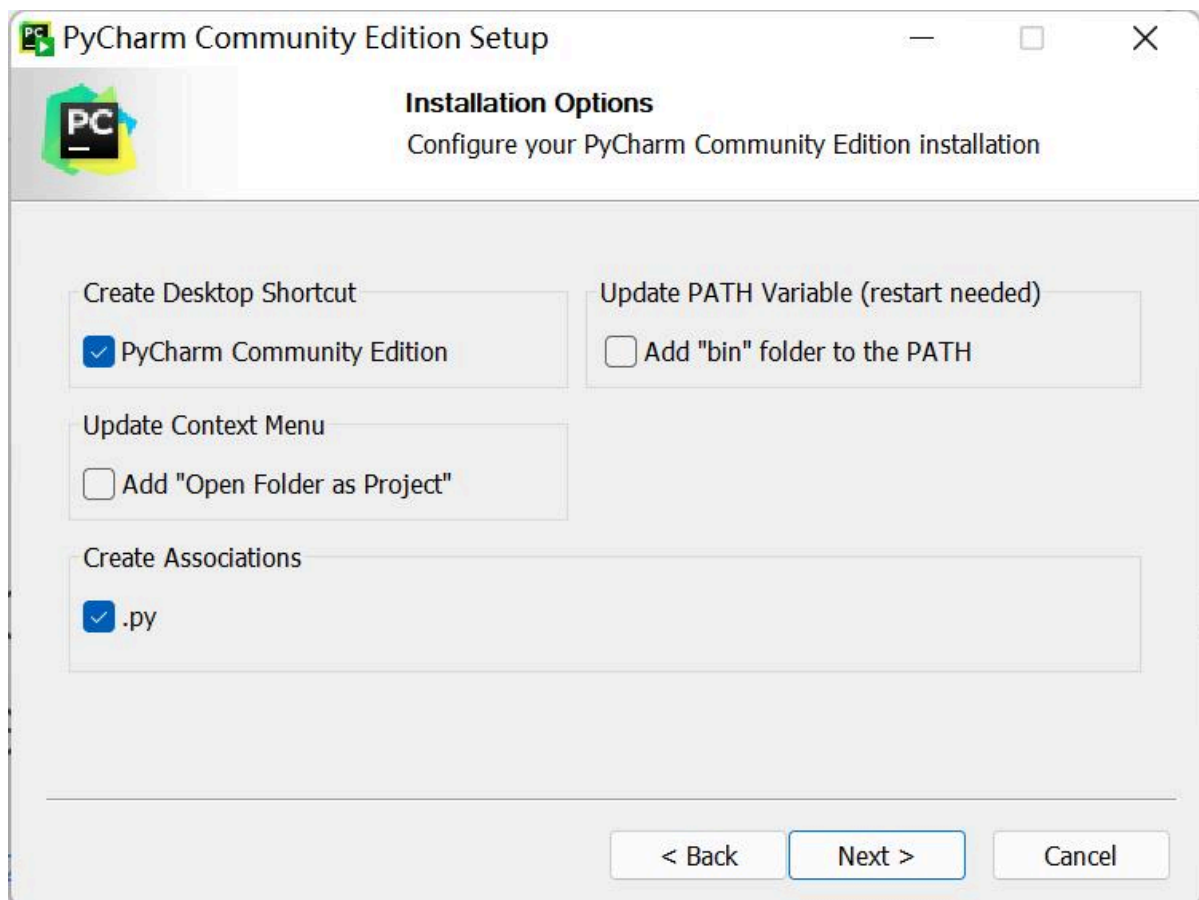
Get the Toolbox App to download PyCharm and its future updates with ease

It is recommended to install the free version.

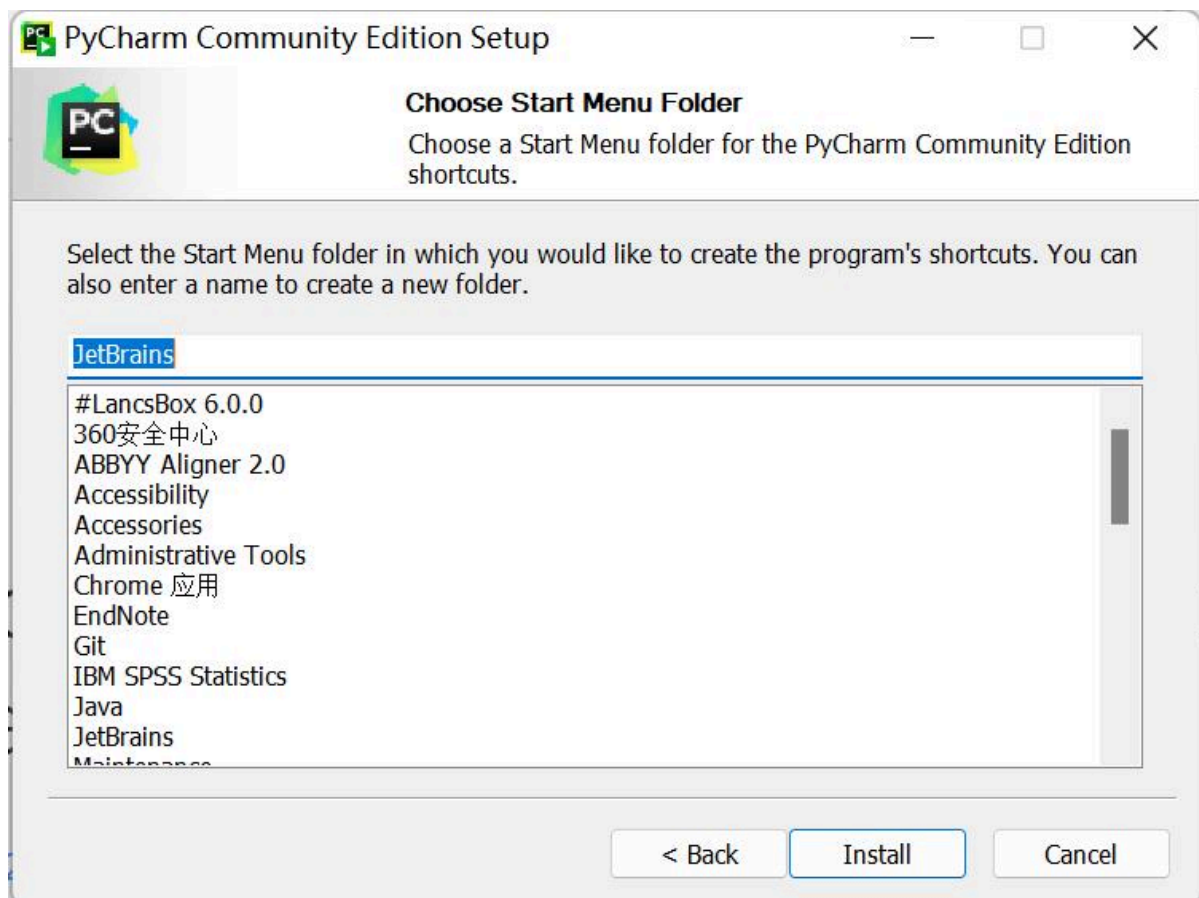
- Click on **Next** :



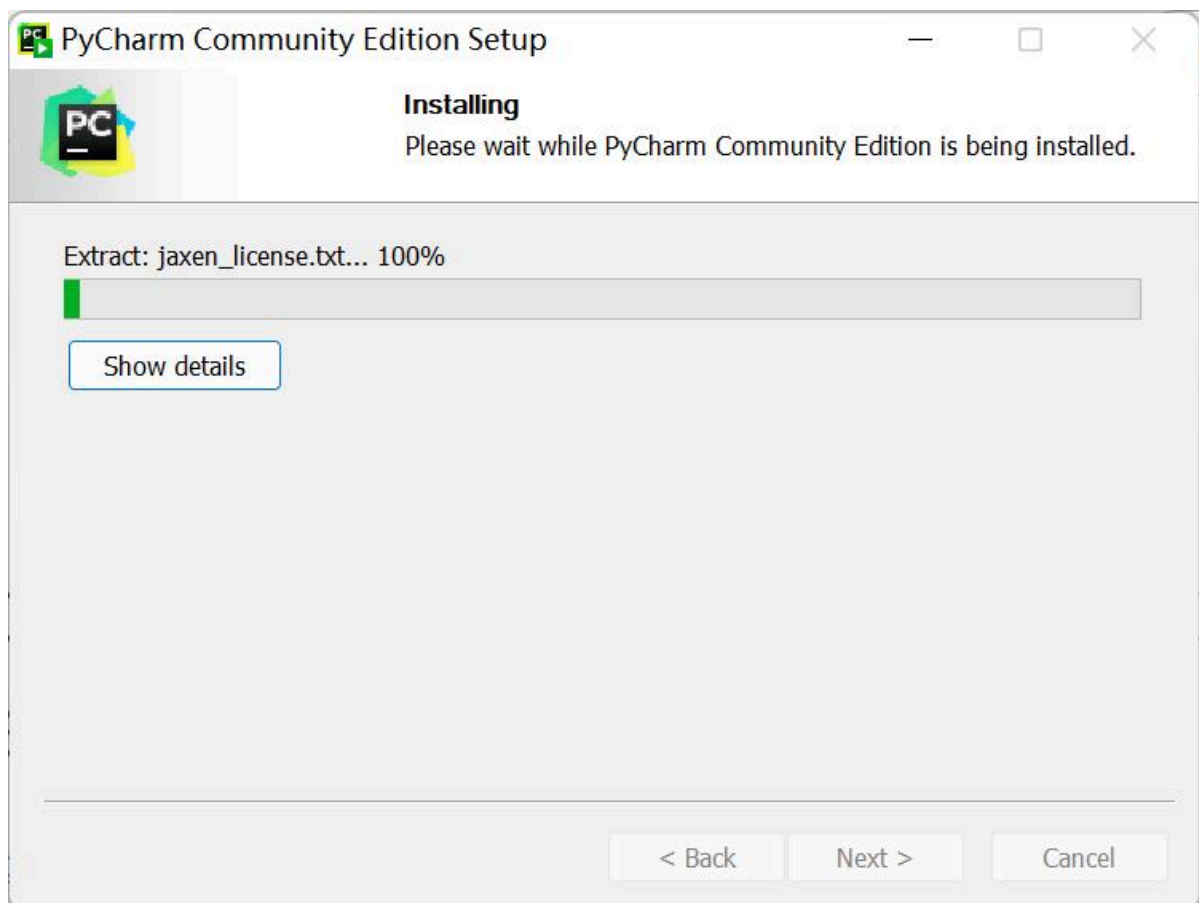
- Select options according to your needs and then select **Next** :



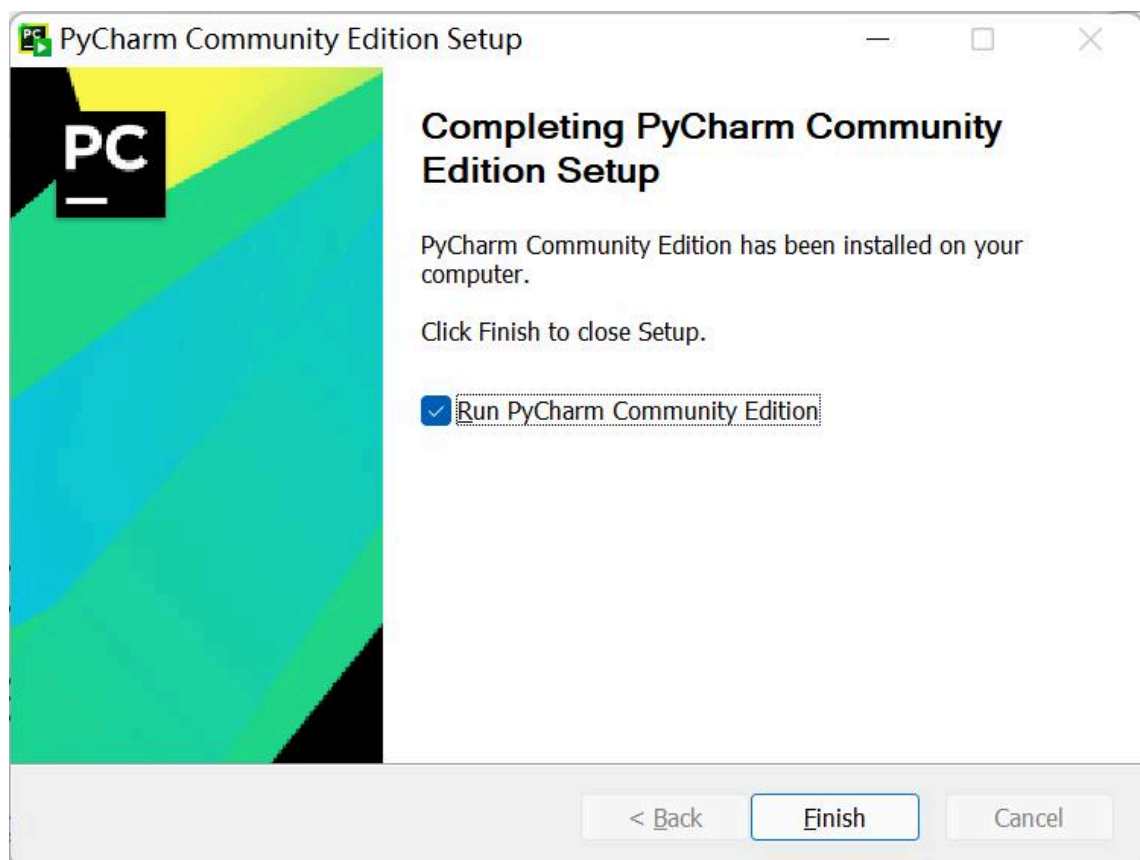
- Tap **Install**:



- Installing:

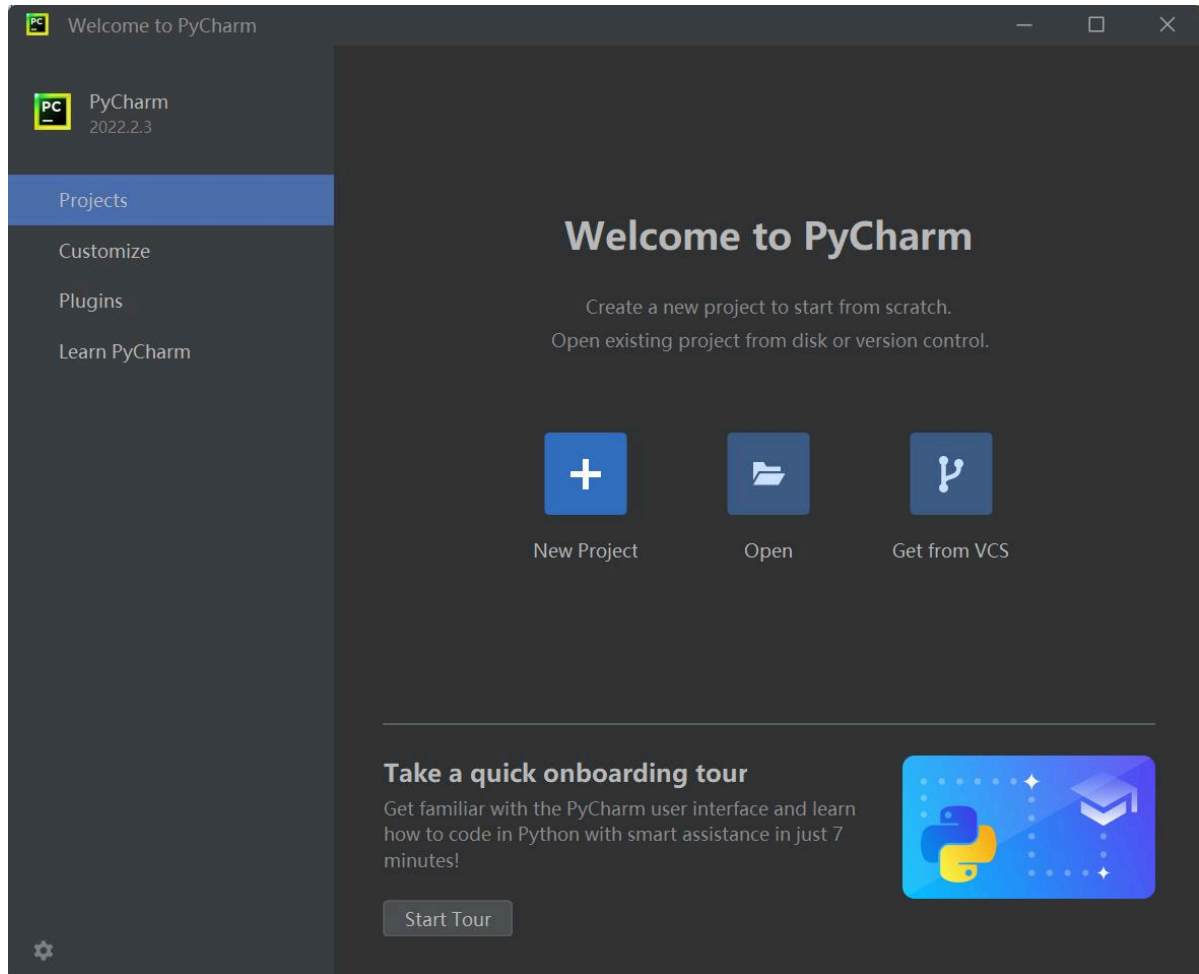


- Tap **Finish**

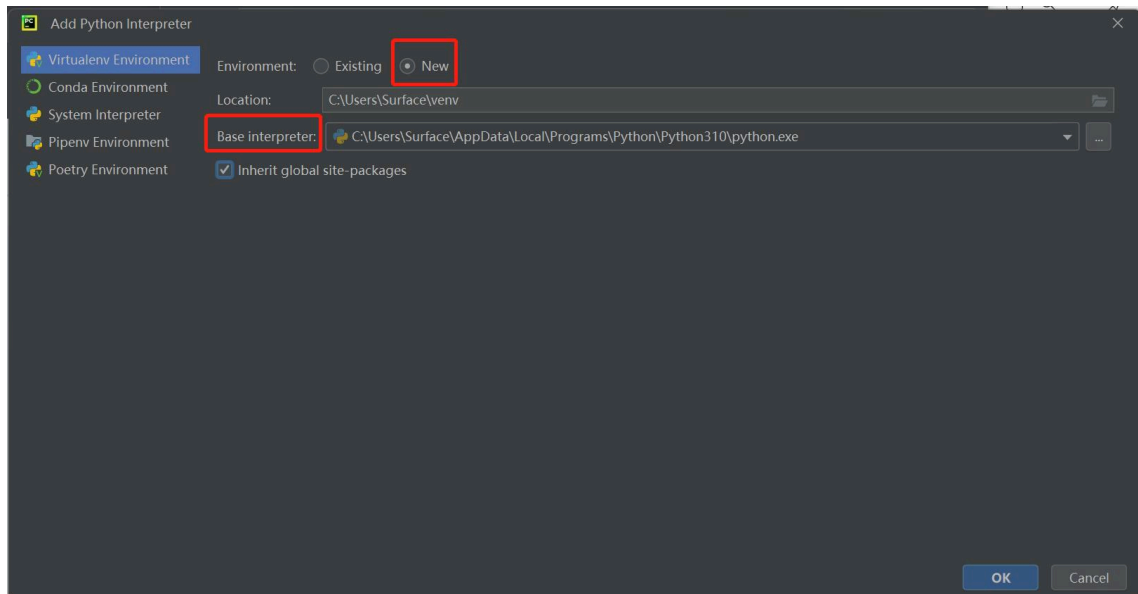
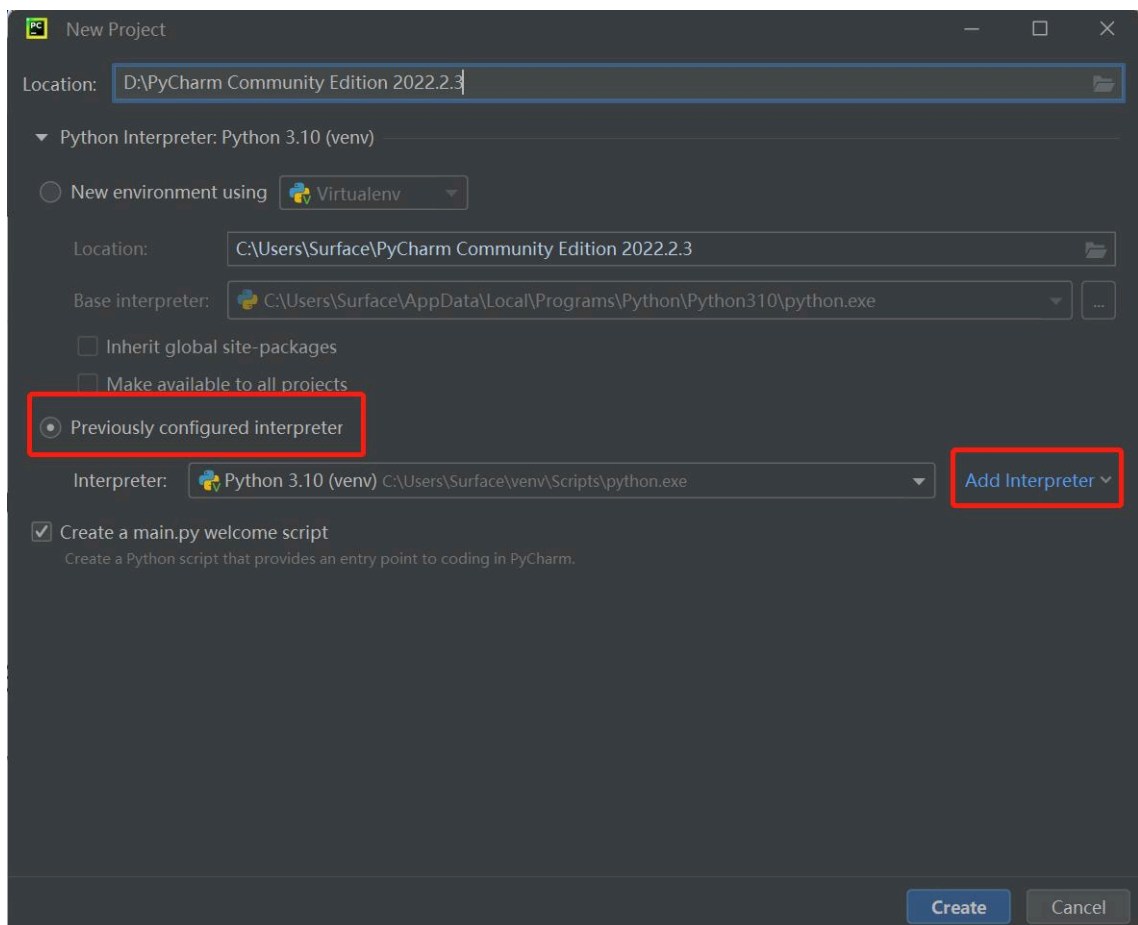


2.2 Create a new project

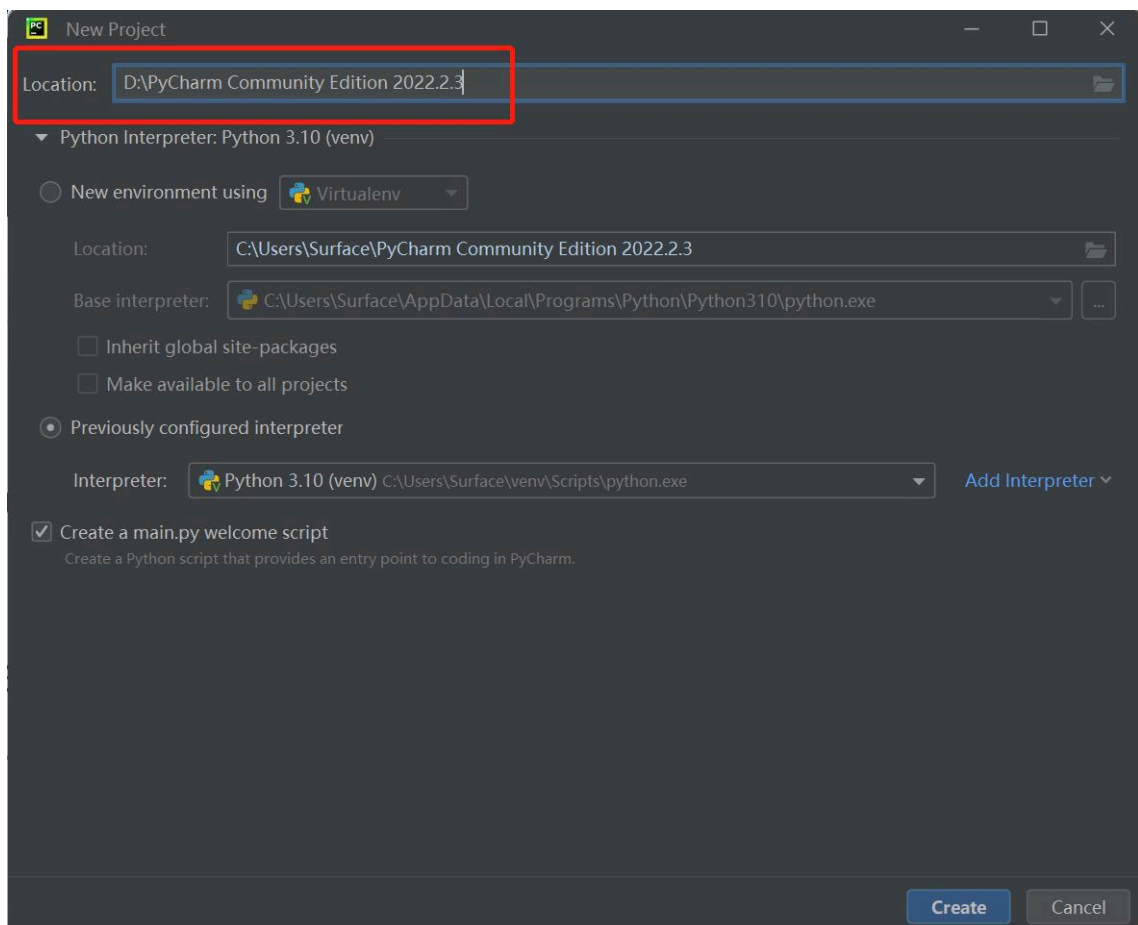
- Click **+New Project** :



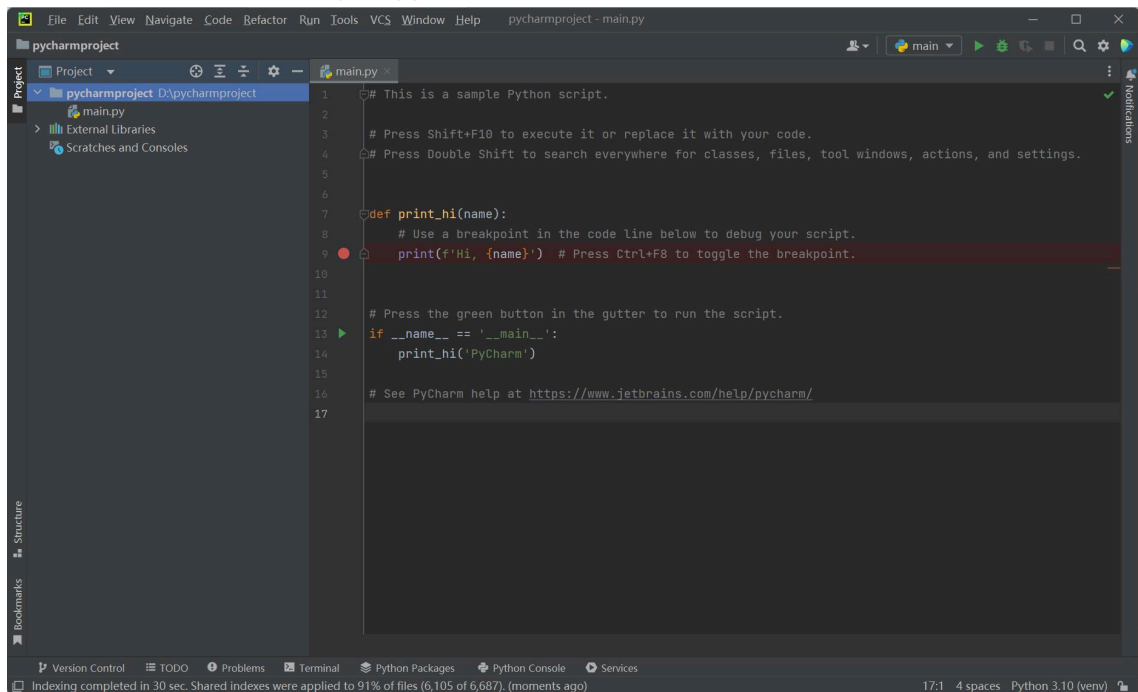
- The **Interpreter** is used to interpret python programs. Select **Add Interpreter** -> **New** to add base interpreter.



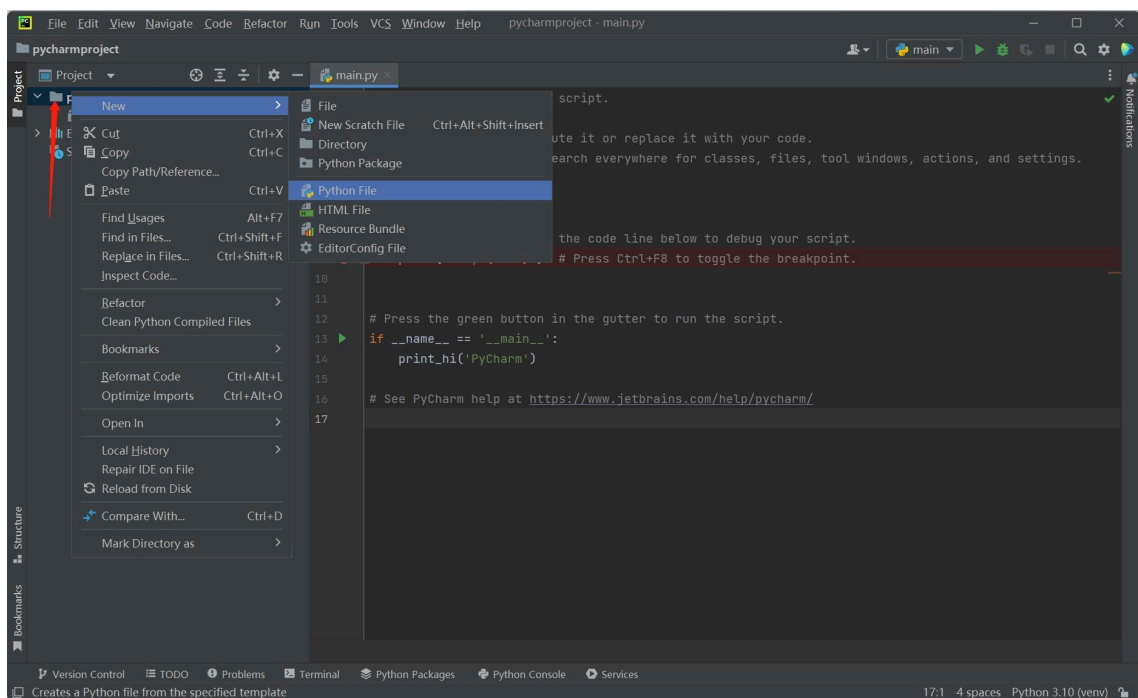
- **Location** refers to the place where to save python file. Choose a file to put your programs.



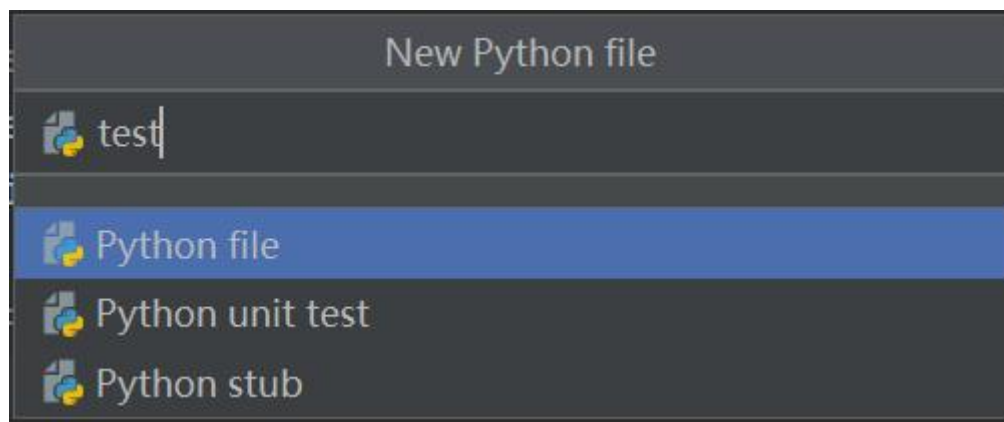
- Click on **Create** and a sample appears:



- Right click on the selection that the red arrow points, and create a new python file.



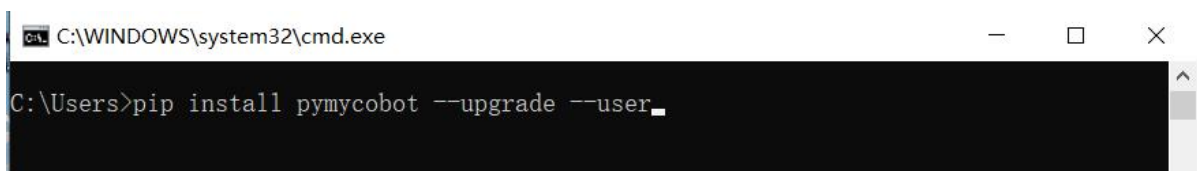
- Type name for the new file.



3 Preparations

- Firmware burning. Firmware serves as a driver for systems to control robots.
 - **M5Stack version** Make sure to burn `mini robot` for **Basic** at the bottom. And then choose `Transponder` function (to receive instructions from Basic), Press `Press A`. `Atom: OK` means connect successfully. Refer to [MyStudio](#) for more information about firmware burning.
- pymycobot installation. Type `pip install pymycobot --upgrade --user` via terminal (Win+R) `cmd` command.

```
pip install pymycobot --upgrade --user
```



- Source code installation. Open a terminal (Win+R, input `cmd`), and type the command below to install.

```
git clone https://github.com/elephantrobotics/pymycobot.git <your-path>
#Fill in your installation address in <your-path>, do not choose the current
default path.
```

```
cd <your-path>/pymycobot
#Go to the pymycobot folder of the downloaded package.
```

```
#Run one of the following commands according to your python version.
```

```
# Install
```

```
python2 setup.py install
```

```
# or
```

```
python3 setup.py install
```

- Update pymycobot

```
pip install pymycobot --upgrade
```

4 Import of pymycobot

This part takes MyCobot 320 M5 as an example to introduce how to control a robot via python.

- Import pymycobot library for MyCobot320 :

```
from pymycobot.mycobot import MyCobot320
```

Notice:

1. If no red wavy line appears below the codes, pymycobot is successfully installed.
2. if a red wavy line appears, got to the address <https://github.com/elephantrobotics/pymycobot> to download pymycobot manually and put it into python library.

